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Constituent of Symbiosis International (Deemed University), Pune

Subject: Computer Vision Laboratory

Experiment - 5

Aim: To segment images based only on color similarity.

Software Tools Required: VS Code, Python 3.12 **Theory:**

1. Introduction

Color-based image segmentation is a process of dividing an image into meaningful regions or objects based solely on the similarity of pixel colors. In this method, the decision of whether two pixels belong to the same region is determined by how close their color values are in a chosen colour space, without considering texture, edges, or spatial features.

2. Concept of Color Similarity

- **Pixels** in an image are represented by their color values.
- If the colors of two pixels are within a certain similarity threshold, they are grouped into the same segment.
- Similarity is usually measured using a distance metric in the chosen color space.

3. Color Spaces Used

Choosing the right color representation is crucial because similarity perception depends on it. Common color spaces include:

1. **RGB (Red, Green, Blue)** – Direct representation but not perceptually uniform.
2. **HSV (Hue, Saturation, Value)** – Better for segmentation because hue represents pure color independent of brightness.
3. **Lab Color Space** – Perceptually uniform, making color difference calculations more accurate.



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4 Segmentation Techniques

Several methods can be applied to segment based on color similarity:

1. Thresholding

- Define a range of color values, and classify pixels falling within that range into the same segment.
- **Example:** Isolating green vegetation in a satellite image by setting hue and saturation thresholds.

2. Clustering Methods

- K-Means Clustering: Groups pixels into k clusters based on color values.
- Mean Shift: Finds dense areas in color space to define clusters automatically.

3. Region Growing

- Start from seed pixels and add neighbouring pixels whose color difference from the seed is below a threshold.

4. Histogram-Based Segmentation

- Use a color histogram to identify dominant colors and assign pixels to corresponding segments.

5. Color Similarity Measurement

To decide if two pixels are similar, a distance metric is used:

Euclidean Distance in RGB or Lab space:

$$D = \sqrt{(R_1 - R_2)^2 + (G_1 - G_2)^2 + (B_1 - B_2)^2}$$

ΔE (Delta E) in Lab space for perceptual difference.

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6. Applications

- Object recognition and tracking
- Medical image analysis (segmenting tissues/organs)
- Agricultural analysis (crop health monitoring)
- Autonomous vehicles (traffic sign detection)

7. Advantages and Limitations Advantages:

- Simple and intuitive.
- Effective when color is the main distinguishing feature.
- Works well for uniformly colored objects.

Limitations:

- Sensitive to lighting changes and shadows.
- Cannot separate objects with similar colors.
- Does not consider texture or shape.

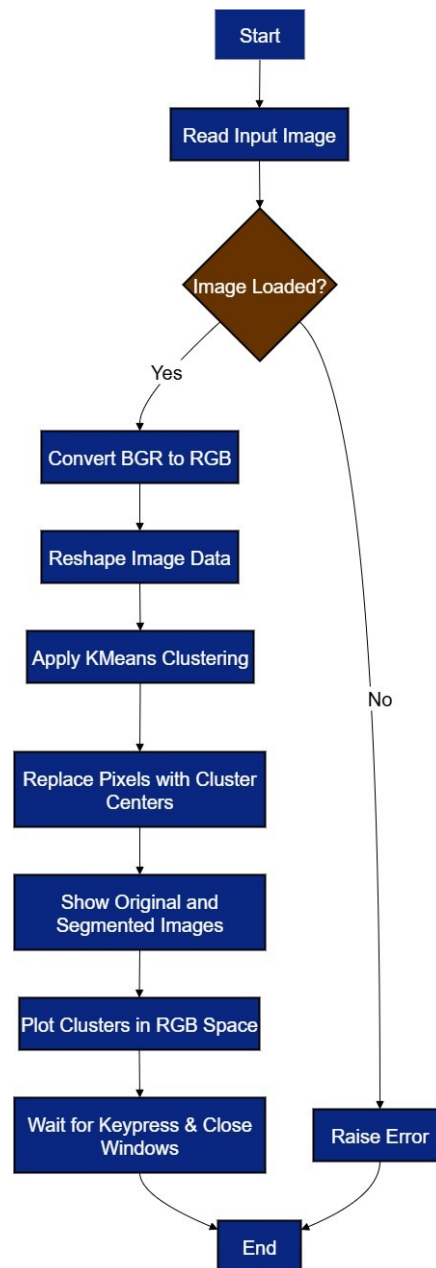
Flowchart

Part-1-RGB Image Segmentation using K-Means

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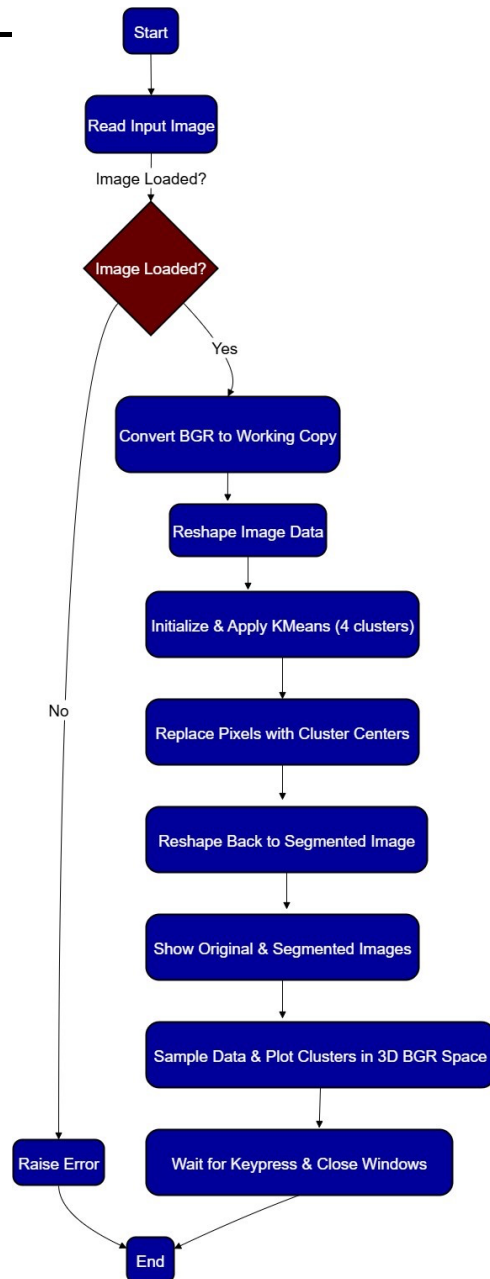


Part-2-BGR Image Segmentation using K-Means

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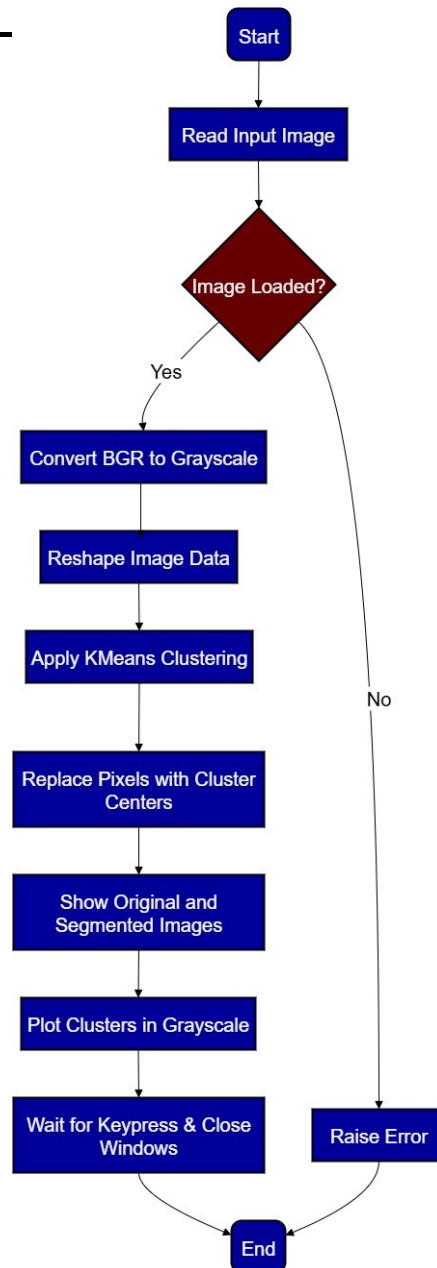


Part-3-Grayscale Image Segmentation using K-Means

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Program with Output:

Program:

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Part-1

```
import cv2
import numpy as np
from sklearn.cluster import KMeans
import matplotlib.pyplot as plt

# Read image
img = cv2.imread('building.jpg')
img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB) # Convert to RGB

# Reshape into 2D array of pixels
data = img_rgb.reshape((-1, 3))

# Apply KMeans clustering
kmeans = KMeans(n_clusters=4, random_state=0, n_init=10)
labels = kmeans.fit_predict(data)

# Replace pixel values with their cluster center colors
segmented_img = kmeans.cluster_centers_[labels]
segmented_img = segmented_img.reshape(img_rgb.shape).astype(np.uint8)

# Show results
cv2.imshow("Original", img)
cv2.imshow("Segmented", cv2.cvtColor(segmented_img, cv2.COLOR_RGB2BGR))
```

```
# Plot KMeans clustering in RGB color space
fig = plt.figure(figsize=(8, 6))
ax = fig.add_subplot(111, projection='3d')
sample = data[:100] # Plot a sample for speed
sample_labels = labels[:100]
ax.scatter(sample[:, 0], sample[:, 1], sample[:, 2], c=sample_labels, cmap='tab10',
ax.set_xlabel('Red')
ax.set_ylabel('Green')
ax.set_zlabel('Blue')
ax.set_title('KMeans Clustering in RGB Color Space')
plt.show()

cv2.waitKey(0)
cv2.destroyAllWindows()
```

Part-2

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```
import cv2
import numpy as np
from sklearn.cluster import KMeans
import matplotlib.pyplot as plt

# Read image
img = cv2.imread('building.jpg')
img_bgr = img.copy() # Already in BGR

# Reshape into 2D array of pixels
data = img_bgr.reshape((-1, 3))

# Apply KMeans clustering
kmeans = KMeans(n_clusters=4, random_state=0, n_init=10)
labels = kmeans.fit_predict(data)

# Replace pixel values with their cluster center colors
segmented_img = kmeans.cluster_centers_[labels]
segmented_img = segmented_img.reshape(img_bgr.shape).astype(np.uint8)

# Show results
cv2.imshow("Original", img)
cv2.imshow("Segmented", segmented_img)
```

```
# Show results
cv2.imshow("Original", img)
cv2.imshow("Segmented", segmented_img)

# Plot KMeans clustering in BGR color space
fig = plt.figure(figsize=(8, 6))
ax = fig.add_subplot(111, projection='3d')
sample = data[:100] # Plot a sample for speed
sample_labels = labels[:100]
ax.scatter(sample[:, 0], sample[:, 1], sample[:, 2], c=sample_labels, cmap='tab10',
ax.set_xlabel('Blue')
ax.set_ylabel('Green')
ax.set_zlabel('Red')
ax.set_title('KMeans Clustering in BGR Color Space')
plt.show()

cv2.waitKey(0)
cv2.destroyAllWindows()
```


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```
import cv2
import numpy as np
from sklearn.cluster import KMeans
import matplotlib.pyplot as plt

# Read image
img = cv2.imread('building.jpg')
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Reshape into 2D array of pixels (single channel)
data = gray.reshape((-1, 1))

# Apply KMeans clustering
kmeans = KMeans(n_clusters=4, random_state=0, n_init=10)
labels = kmeans.fit_predict(data)

# Replace pixel values with their cluster center values
segmented_img = kmeans.cluster_centers_[labels]
segmented_img = segmented_img.reshape(gray.shape).astype(np.uint8)

# Show results
cv2.imshow("Original Gray", gray)
cv2.imshow("Segmented Gray", segmented_img)
```

```
# Replace pixel values with their cluster center values
segmented_img = kmeans.cluster_centers_[labels]
segmented_img = segmented_img.reshape(gray.shape).astype(np.uint8)

# Show results
cv2.imshow("Original Gray", gray)
cv2.imshow("Segmented Gray", segmented_img)

# Plot KMeans clustering in grayscale
plt.figure(figsize=(8, 4))
sample = data[:100].flatten() # Sample for speed
sample_labels = labels[:100]
plt.scatter(np.arange(len(sample)), sample, c=sample_labels, cmap='tab10', s=10)
plt.xlabel('Pixel Index (sampled)')
plt.ylabel('Gray Value')
plt.title('KMeans Clustering in Grayscale')
plt.show()

cv2.waitKey(0)
cv2.destroyAllWindows()
```

Output:

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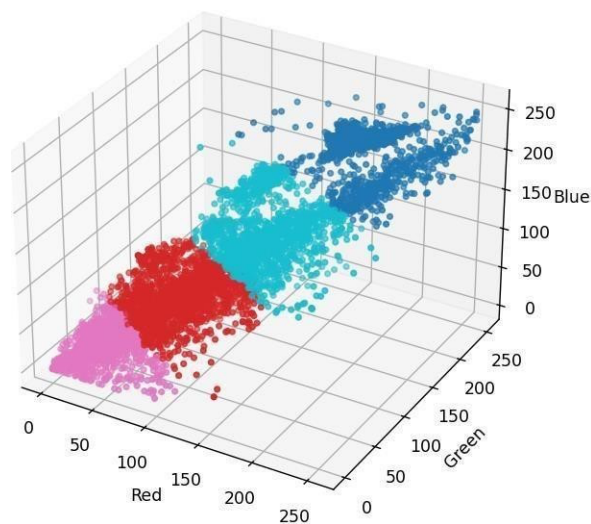


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Part-1



KMeans Clustering in RGB Color Space

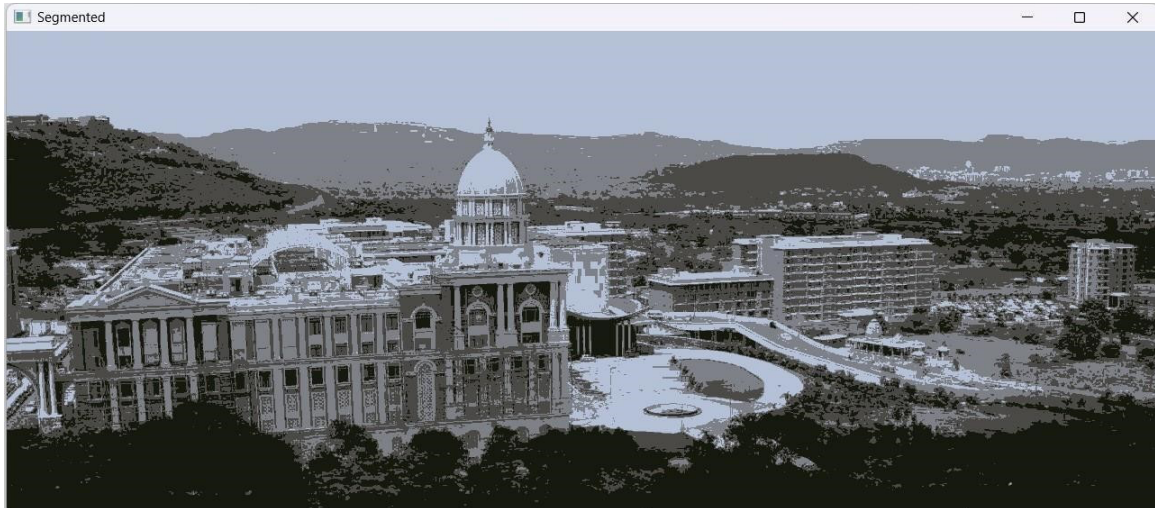


Part-2

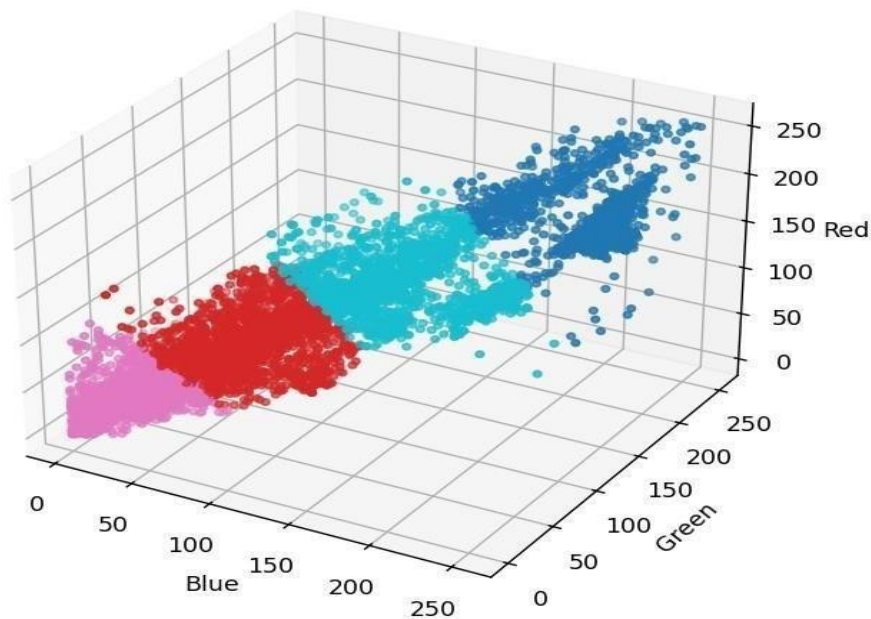
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KMeans Clustering in BGR Color Space



Part-3

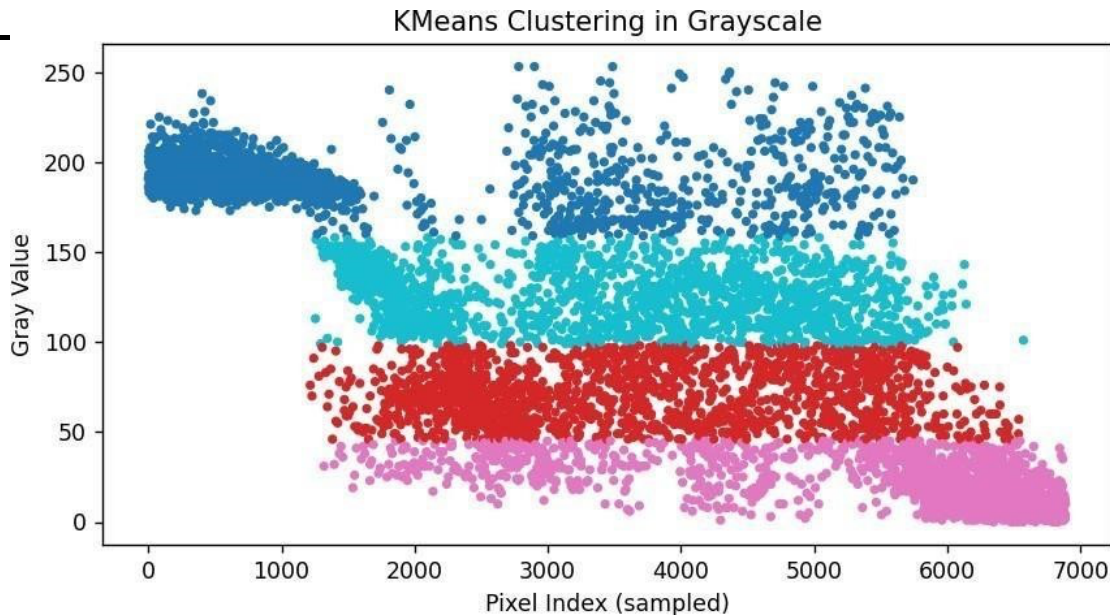
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Conclusion/Inferences:

The implementation demonstrates how K-Means clustering can effectively segment images in different colour spaces—RGB, BGR, and Grayscale—based on pixel similarity.

- In RGB and BGR modes, segmentation groups pixels by their colour components, producing visually distinct regions that highlight dominant colours.
- In Grayscale mode, segmentation is based solely on intensity levels, simplifying the image into separate brightness regions.

This approach successfully reduces image complexity while preserving key structural details, making it suitable for applications such as object detection, image compression, and feature extraction. However, its performance can be influenced by the choice of k clusters and variations in lighting.